

QCC Labs OSINT

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Goal: Learn some OSINT fundamentals and about Tsurugi

Pre Requirements:

- Personal laptop
- Preferable:
    - Download and install Virtual Box ahead of time.
    - Download and boot Tsurugi OVA file ahead ahead of time will allow you to follow on from the start instead of having to download and import the VM.
    - At the time of writing this lab Tsurugi requires just under 90 GB of disk space. You can substitute Kali if you want as many of the tools are pre-installed.
    - You may need to temporarily disable your Windows Firewall.

OSINT, or Open Source Intelligence, is the process of collecting and analyzing publicly available information from sources like the internet, social media, public records, and news to produce actionable intelligence. It is a legal and ethical method of gathering intelligence without engaging in unauthorized activities. This information is then used for various purposes, including national security, law enforcement investigations, competitive business analysis, and cybersecurity.

Key aspects of OSINT

- Definition: The systematic collection and analysis of data that is freely available to the public.
- Purpose: To derive meaningful conclusions and make informed decisions by uncovering patterns, threats, or opportunities.
- Legal and ethical: It relies on data that is accessible without illegal or unauthorized methods.

Sources of OSINT

- Internet: Websites, blogs, online forums, and news outlets.
- Social Media: Platforms like Facebook, X (formerly Twitter), LinkedIn, and YouTube.
- Public Government Data: Government reports, budgets, public records, and speeches.
- Academic and Professional Publications: Research papers, journals, and conference proceedings.

- Commercial Data: Business and financial assessments, and commercial imagery.
- Grey Literature: Technical reports, patents, and unpublished works.

Applications of OSINT

- National Security: Providing intelligence to policymakers and national security officials.
- Law Enforcement: Investigating crimes, locating missing persons, and tracking criminal networks.
- Cybersecurity: Identifying vulnerabilities in networks and systems by monitoring public information about threats.
- Business: Gaining competitive intelligence, performing market research, and tracking brand sentiment.

It is important that we are teaching these skills to help you protect yourself, your friends, your co-works, and your organization. Using these tools, techniques, and tactics can very quickly cross the line from completely acceptable to creepy. At worse when combined with actions in the real world can have consiquences such as stalking, harassment, liable, slander, intimidation and much worse.

In the immortal words of Ron Burgundy, “You stay classy, San Diego”.

> “
> Your DFIR Linux distribution
>
> Tsurugi Linux is a DFIR open source project that is and will be totally free, independent, without involving any commercial brand
>
> Our main goal is share knowledge and "give back to the community"
> ”

https://tsurugi-linux.org/

Tsurugi Linux also has a bunch of OSINT Tools built in.

Here is a list of all the tools included with Tsurugi 2024:
https://tsurugi-linux.org/documentation_tsurugi_linux_tools_listing_2024.php

Thank you to our friends at Tsurugi for continuing to support the community!

Next we need a hypervisor. Here we get to choose but two great options are VMWare and VirtualBox. VMWare was recently purchased by Broadcom and when that happened the VMWare Workstation Pro (Windows & Linux) and Fusion Pro (Mac) product went back behind a paywall for personal use. After a bit of a backlash for this and other changes to the VMWare line Broadcom once again restored your ability to use Workstation Pro and Fusion for free under personal use, you just to make a Broadcom account. VirtualBox is Open Source solution that is owned and supported by our friends at Oracle.

While my intention was for you to choose your favorite hypervisor and use it I wasted 90 minutes trying and failing to get the free personal use licenses from VMWare. Thus, I strongly suggest using virtual box! You can read and follow along my mis-adventures with Broadcom at the end of the Project ADHD Lab.

Remember to always read the terms of service for any open source tool. Generally it is cool to use these tools in advancement of personal knowledge and skill. However, the corporate use of these tool might have some restrictions. In general if we like / loves the tools that we use, it is just good form to support the developers.

Legal Disclaimer:

Before we go any further, it is critical to only use these tools on device you either own or have been given explicit (preferably written) authorization to perform these explicit acts on. The Computer Fraud and Abuse Act is a very real and serious thing. We all here at QCC want you to stay on the correct side of prison, the outside.

Prerequisite:

Download VirtualBox:

1. Go to: https://www.virtualbox.org/ and click downloads
2. Select your distribution to start your download
3. So much simpler than VMWare

Install Virtual Box:

1. Go to download location and double click. Accept UAC pop up.
2. Please note that exact install options might change as app is updated. This guide was created 8/24/2024.
3. On the first generic install screen click next.
4. The next screen allows you to customize your Virtual Box install. I kept with the default and clicked next.
5. Click Yes to the network interruption message.
6. Click yes if you get any missing dependencies.
7. At the next customization screen, I kept the default.
8. Click Install on the 'Ready to Install Screen'
9. Click Finish

Make sure that you are connected to the internet.

Now we need to get the Tsurugi free Virtual Appliance. Visit this site and click the free button.
https://tsurugi-linux.org/downloads.php and click on one of the mirrors and click on the *vm.ova file

a copy of this file will also be loaded onto the lab wifi.



Mirror ftp.nluug.nl - Tsurugi Linux project

01. Tsurugi Linux [Lab]

tsurugi_linux_2024.1.iso	30-Jul-2024 14:31	16.7G
tsurugi_linux_2024.1_vm.ova	30-Jul-2024 14:31	33.7G

02. Tsurugi Linux Acquire

tsurugi_acquire_2021.1.iso	09-Apr-2021 14:03	1.1G

03. Bento toolkit

bento_2025.8.7z	25-Aug-2025 17:46	614M

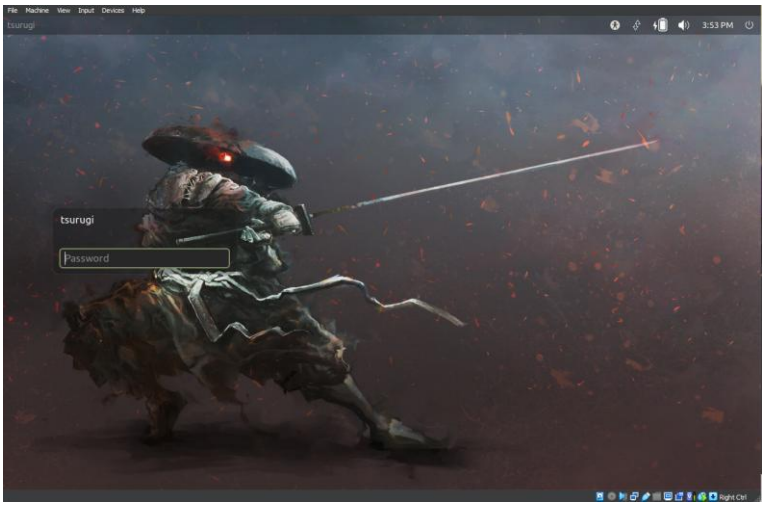
Tsurugi Linux OVA into VirtualBox as the Hypervisor

Now we need to load the Tsurugi Virtual Appliance into the Virtual Box:

1. As you will need you host system and your VM to have an IP address so that they can communicate with each other connect to a wifi (wither QCCLabs or your cell phone hotspot as in this case you will not be in in conflict with the Computer Abuse and Fraud Act)
2. Open Virtual Box
3. In the Virtual Box Console click ‘File -> Import Appliance’ or press ‘Control + I’
4. Click the icon of the folder with an up arrow on the right of the import wizard and point to the OVA file you just downloaded. Click ‘OK’ to populate the File location. Once the File location is populated click ‘Next’.
5. Review the settings.
    a. Double click on the name and change it to Tsurugi if it does not auto populate.
    b. Double click on CPU to change the number of cores you give your VM. (min requirements at time of writing this is 4 cores)
    c. Double click on the ram number to allow you to change it to at least 5120 MB. (min per documentation)
    d. Adjust the ‘Machine Base Folder’ if you want you VM in a different place.
    e. Click Finish
    f. You might be asked to agree the Software License Agreement
6. You can watch the progress as Virtual Box imports the VM.
7. Once the Tsurugi Virtual Appliance has been imported. Either double click the Tsurugi VM or apply a single click to Tsurugi VM and then click start. If you get an error like I did we need to do a few more steps.
    a. Click on the VM and then the golden Settings gear
    b. Click on the Networking icon in the Settings pop up.
    c. Ensure you in the Adapter 1 tab
    d. The official Tsurugi instructions say you need to select the proper Networking Name under the ‘Name’ field. I had to go in and select the correct name of my network card to make it work.

8. Now try to power on your system again. I know apologize for the vagueness of ‘play with the network settings until you resolve the issue with a virual appliance that someone else made’. But, we are hackers! Sometime we will relentless bash our head into a technical issue until we get good. It is like the Poet Laureate Chumbawamba once sang, “Oh, Danny Boy, Danny Boy, Danny Boy, I get knocked down, But I get up again, You're never gonna keep me down, I get knocked down, But I get up again, You're never gonna keep me down, I get knocked down, But I get up again, You're never gonna keep me down, I get knocked down, But I get up again, You're never gonna keep me down”

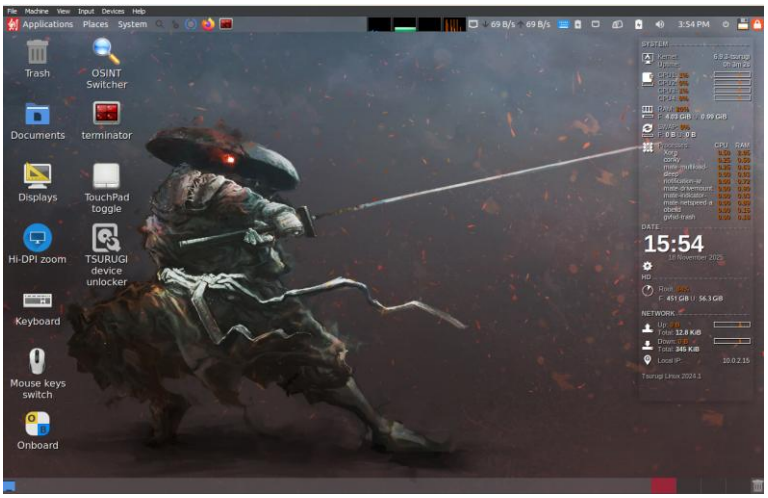
Initial Boot of Tsurugi Virtual Appliance



If you use the OVA file the credentials will be:

U: tsurugi

P: tsurugi



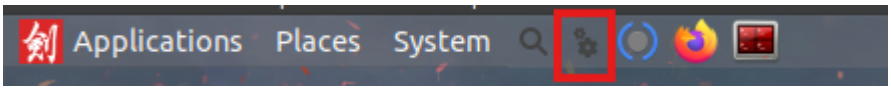
In the real world you might just want to change the password and apply all updates. (The updates may take a long time)

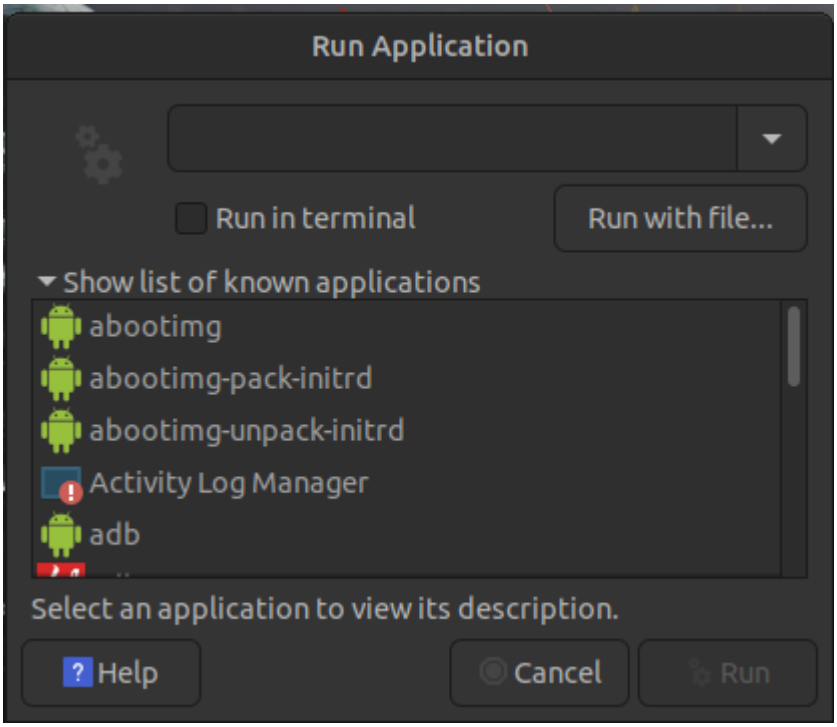
If the network does not work shut down the VM. Then right click on the VM -> Setting -> Network and switch the attached to from NAT to bridge. (Please not you might also have to set your active network card). Power up Tsurugi and try again.

Please note Tsurugi has the proclivity of grabbing your mouse and locking it to the VM. Simply tap the right Control button on you keyboard to return your mouse to the host VM.

To access most of the tools we will use click on ‘Applications’ in the upper left -> ‘Tsurugi’ -> ‘OSINT’

You can also click the icon of the two gears above the head of the battle monk you can simply type the name of the application you want to run





Google Dorks

Open up Firefox and do a quick web searchto ensure your host can connect to the internet..

‘Applications’ -> ‘Internet’ -> ‘Firefox’

Google dorks are meticulously crafted search parameters that utilize Google's advanced operators to find specific types of information on the web. The goal is to make your searches more pinpoint and accurate.

These commands can be ‘and’ / ‘or’ together to refine your search

Google Dork	Description	Example
site:	Restricts a search to a specific website.	site:example.com login

filetype:	Searches for a specific type of file, such as a PDF or a SQL file.	filetype:pdf report
inurl:	Finds pages with a specific word or phrase in the URL.	inurl:admin
intitle:	Locates pages with a specific word or phrase in the page title.	intitle:"index of"
intext:	Finds pages that contain specific words within the body of the page.	intext:"sensitive information"
link:	Finds pages that link to a specific URL.	link:example.com
ext:	Similar to filetype:, but searches for a specific file extension.	ext:doc OR ext:xls
cache:	Shows the cached version of a webpage.	cache:example.com

From Firefox go to www.google.com and search for 'Pomsky videos'. Notice you get a mix of sites from TikTok, Youtube, Facebook, Rumble and more.

Now let's refine our search to 'Pomsky videos site:youtube.com'. now you see that we are just getting youtube files.

Now lets search 'site:uat.* * inurl:login' this is interesting as we get a ton of UAT downloads. This would be of interest for me to help understand what UAT environments are publicly easily discoverable to the internet.

Likewise we can use 'site:uat.* * filetype:pdf' to return UAT sites that Google has scanned a PDF file.

As you can see this is a very powerful tool that can help you professionally and personally!

For more Google Dorks you can view exploit -db.com list here:

https://www.exploit-db.com/google-hacking-database

Fast-Google-Dork-Scan

Now that we have shown you the hard way lets look at the easy way with the built in Fast-Google-Dorks Scan

> "
>
> Fast Google Dorks Scan
> The OSINT project, the main idea of which is to collect all the possible Google dorks search combinations and to find the information about the specific web-site: common admin panels, the widespread file types and path traversal. The 100% automated.

“

https://github.com/IvanGlinkin/Fast-Google-Dorks-Scan

‘Applications’ in the upper left -> ‘Tsurugi’ -> ‘OSINT’ -> 'Web' -> 'Fast-Google-Dork-Search'

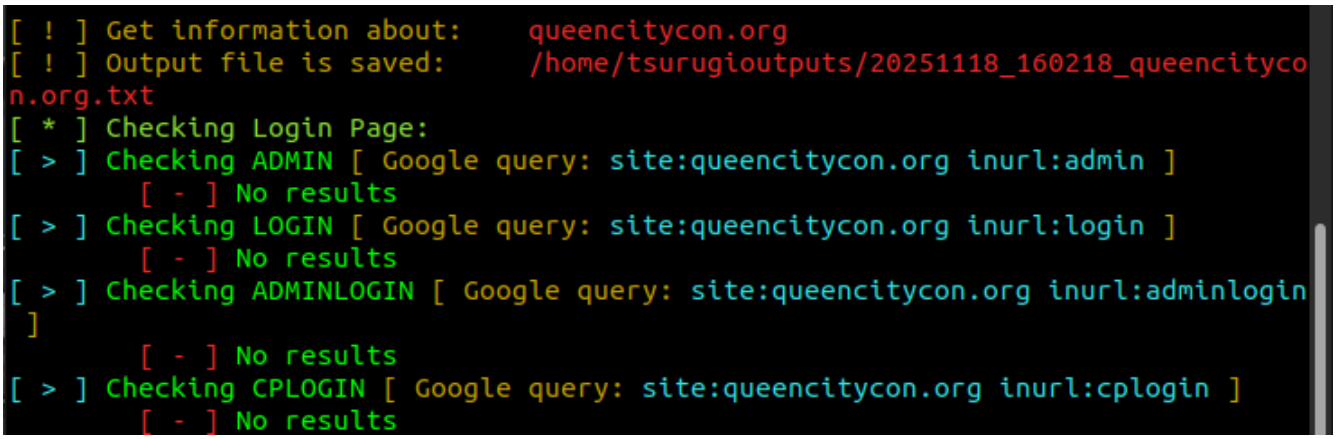
You can see that they will give us an example of :



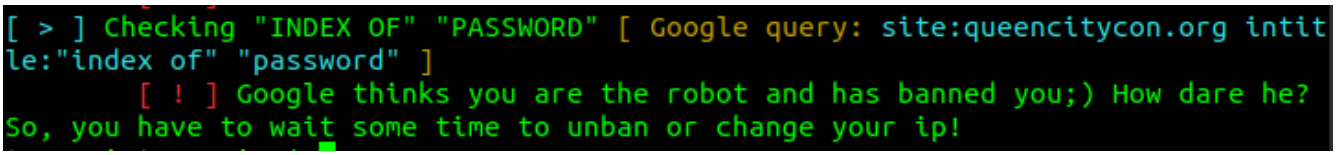
Simply copy and past the command but replace the URL with ‘queencitycon.org’ and hit enter

bash /opt/Fast-Google-Dork-Scan queencitycon.org

You will see that the it will start the scan and tell you where the Output file is sent. Please note our site is very small but it did take a few minutes to run



You can play with some of the scan settings so that your network infrastructure does not think you are a robot. But hopefully you see the power of this tool!



H8mail

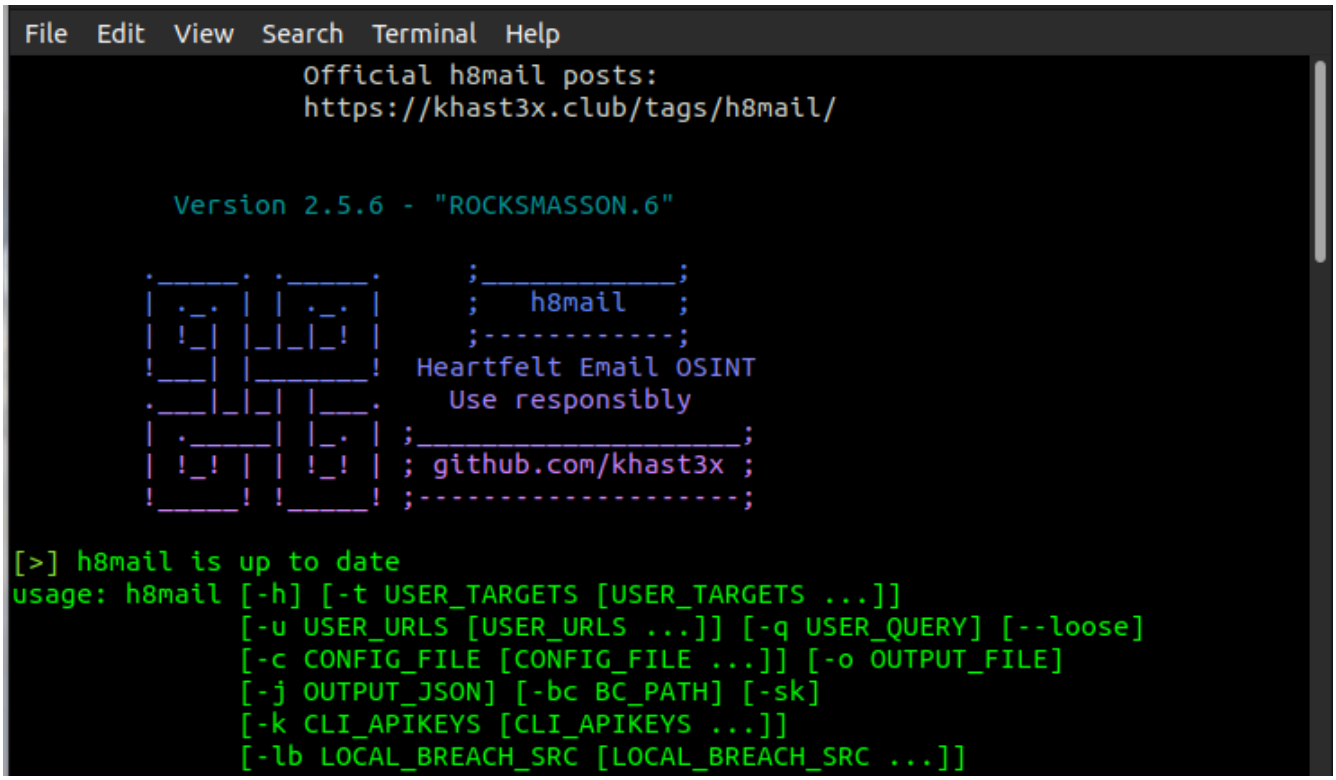
“

h8mail is an email OSINT and breach hunting tool using different breach and reconnaissance services, or local breaches such as Troy Hunt's "Collection1" and the infamous "Breach Compilation" torrent.

“

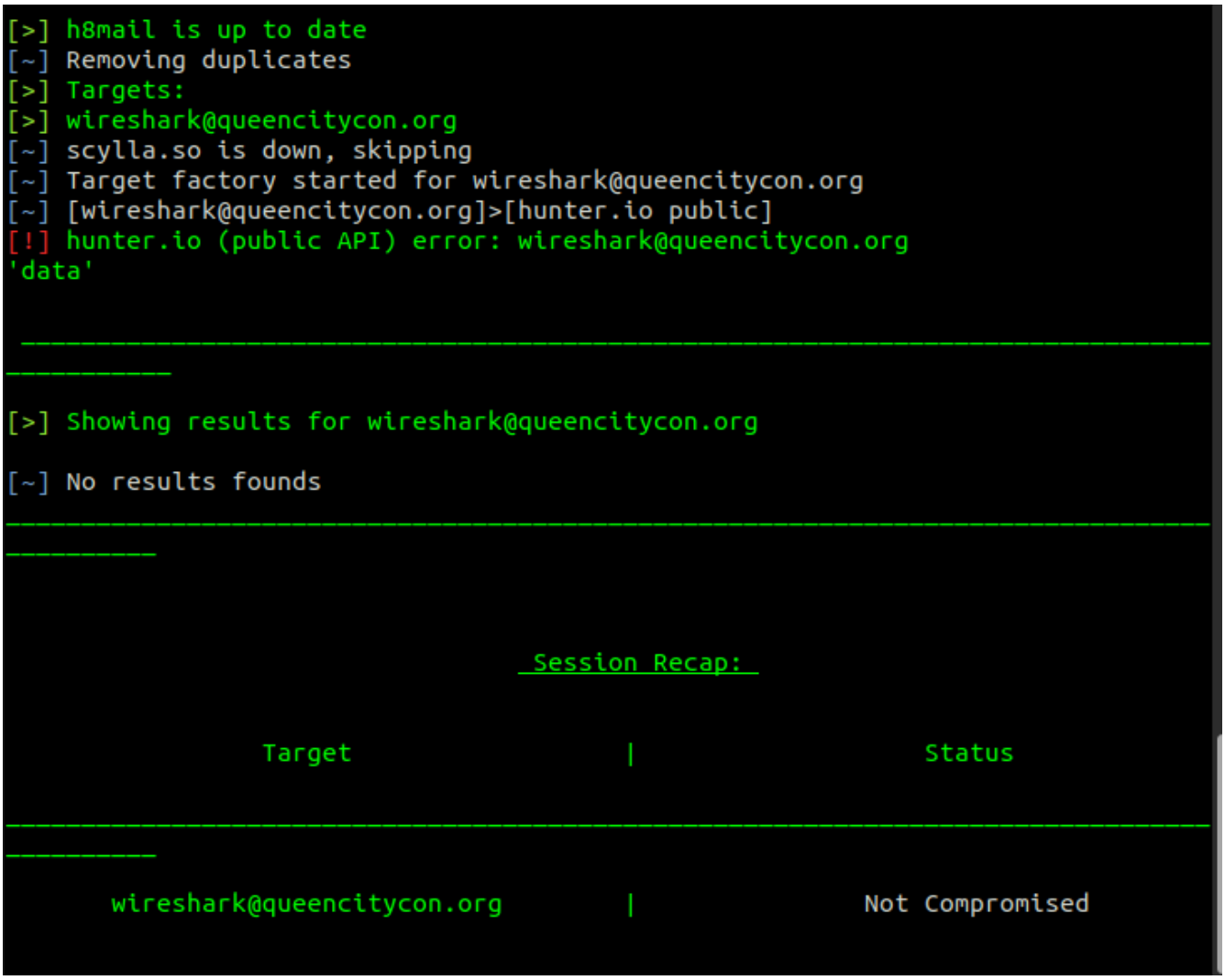
https://github.com/khast3x/h8mail

To open h8mail click ‘Applications’ in the upper left -> ‘Tsurugi’ -> ‘OSINT’ -> 'Emails' -> 'h8mail' to open up the instructions string.



To look for a single email

h8mail -t wireshark@queencitycon.org



At least at the time of the writing of this lab The Wireshark's email is NOT compromised.

You can check h8mail's documentation for more advanced queries that allow you more advance options such as searching for multiple IPs at a time, search against a defines set of targets.

Photon

"

Key Features
Data Extraction
Photon can extract the following data while crawling:

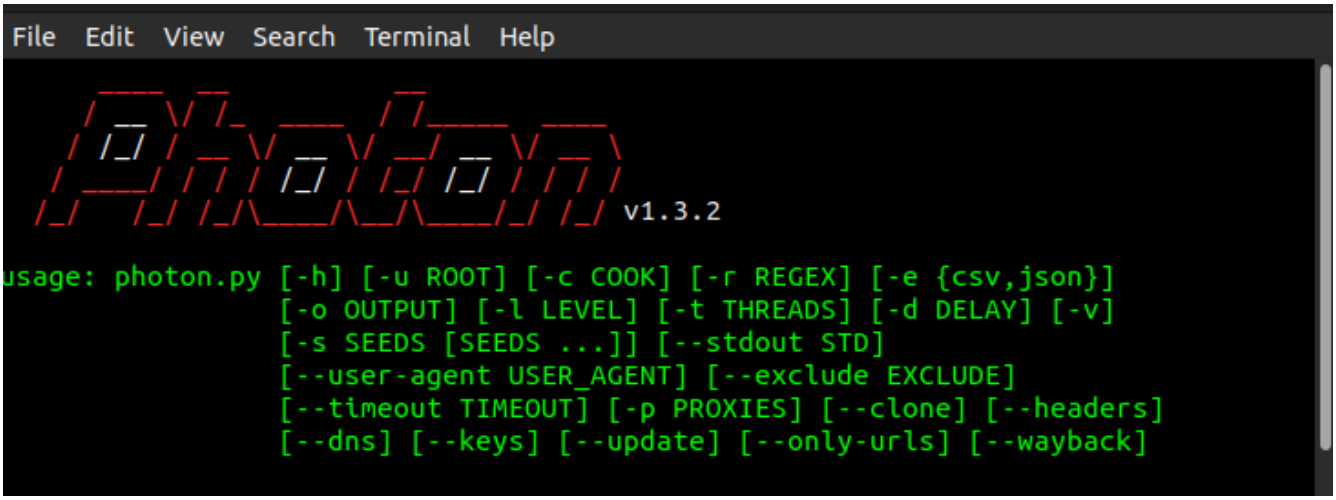
- URLs (in-scope & out-of-scope)
- URLs with parameters (example.com/gallery.php?id=2)
- Intel (emails, social media accounts, amazon buckets etc.)
- Files (pdf, png, xml etc.)
- Secret keys (auth/API keys & hashes)

- JavaScript files & Endpoints present in them
- Strings matching custom regex pattern
- Subdomains & DNS related data
- The extracted information is saved in an organized manner or can be exported as json.

“

https://github.com/s0md3v/Photon

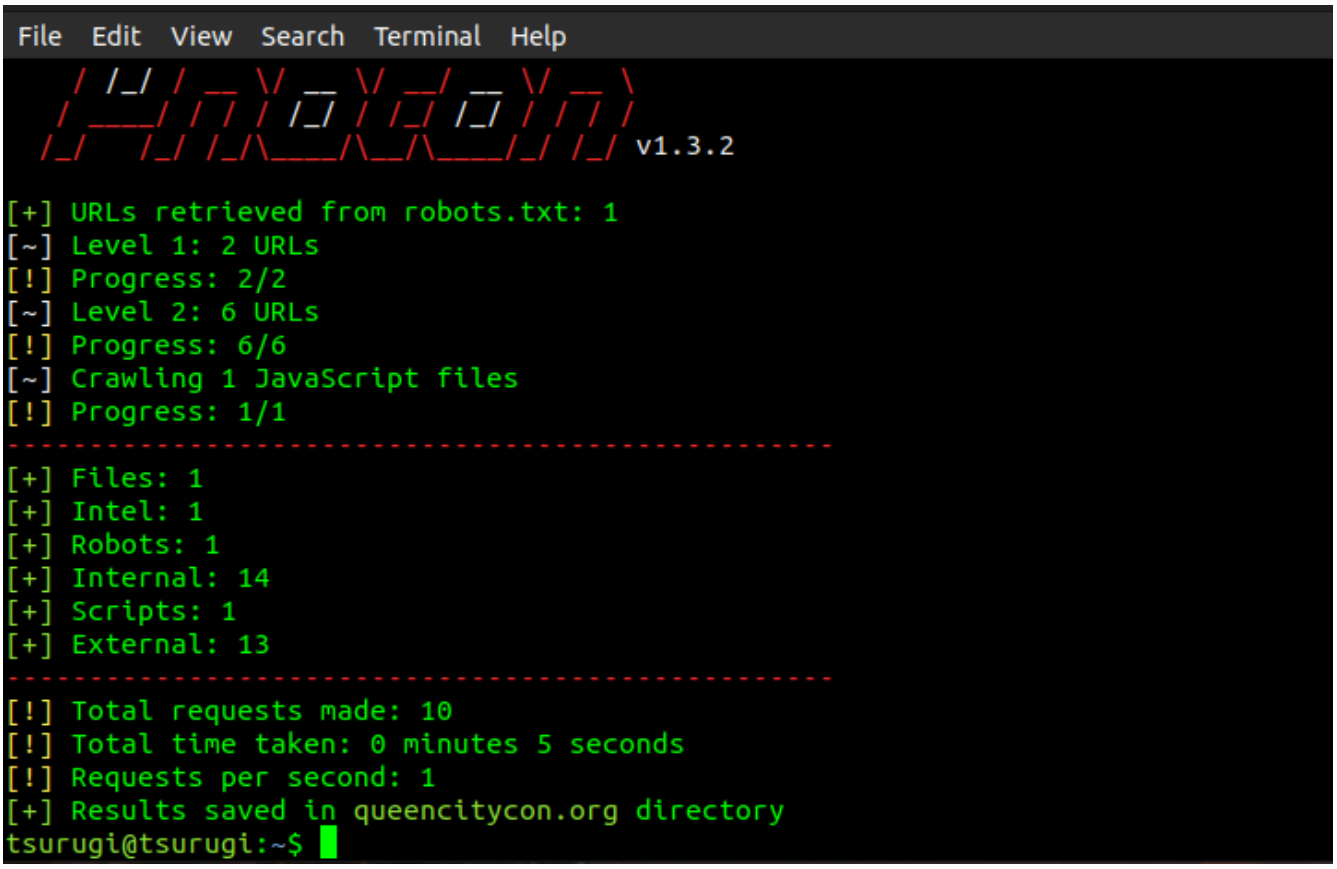
To open the tool click ‘Applications’ in the upper left -> ‘Tsurugi’ -> ‘OSINT’ -> 'Web' -> 'Photon' and you will be greated with the instruction stream.



Lets run a basic scan:

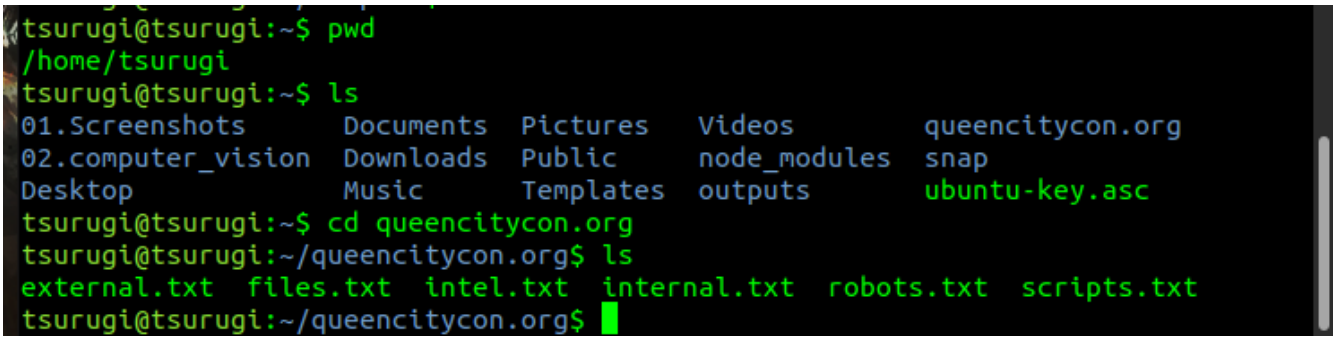
photon.py -u queencitycon.org

At the time of the writing of this lab here is the basic results



As you can see results are saved in a folder under the 'Present Working Directory' cd

To see your 'Present Working Directory' simply use the command pwd



You can cd into the folder you created and do a 'ls' to see the default files.

You can either use a program like 'cat', 'nano', 'vim', and others to read the files directly in the source folder or click 'Document' icon -> 'Home Folder' -> the folder that bears the name of URL you scaned.

Either way look into a few of the file results.

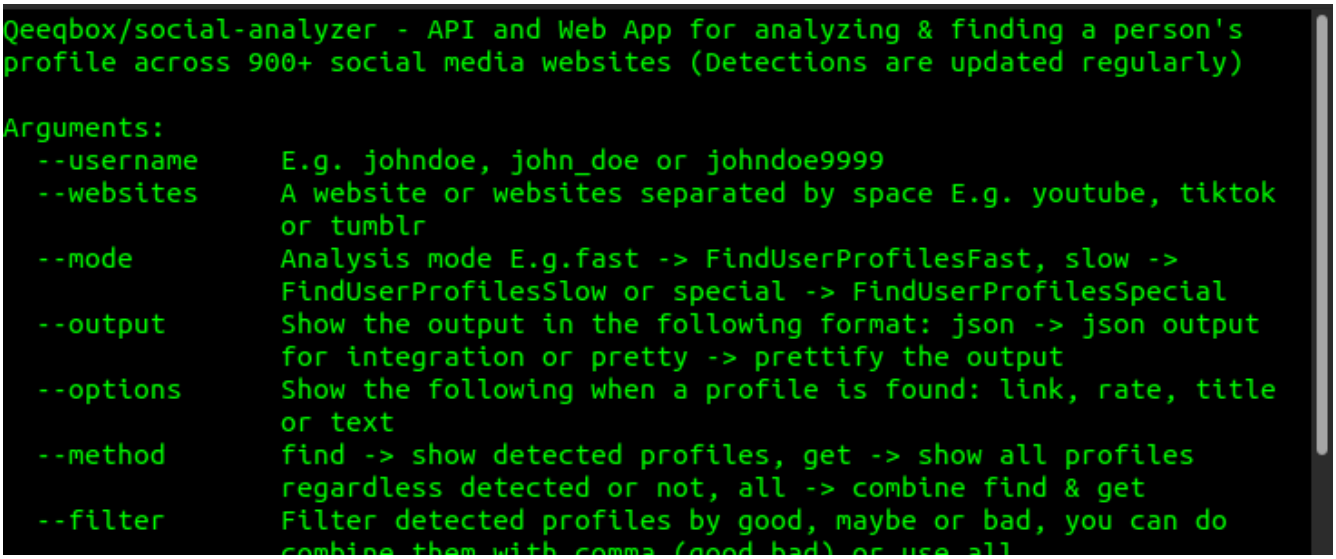
You can check instruction when you start the application or the instruction on Phantom's github for advanced options:
https://github.com/s0md3v/Photon

social-analyzer

> “
> Social Analyzer - API, CLI, and Web App for analyzing & finding a person's profile across +1000 social media \ websites. It includes different analysis and detection modules, and you can choose which modules to use during the investigation process.
>
> The detection modules utilize a rating mechanism based on different detection techniques, which produces a rate value that starts from 0 to 100 (No-Maybe-Yes). This module is intended to have fewer false positives.
> The analysis and public extracted information from this OSINT tool could help investigate profiles related to suspicious or malicious activities such as cyberbullying, cyber grooming, cyberstalking, and spreading misinformation
> “

https://github.com/qeeqbox/social-analyzer

To open click ‘Applications’ in the upper left -> ‘Tsurugi’ -> ‘OSINT’ -> 'Social' -> 'social-analyzer'



Once again we have a very helpful instruction set.

Form here we can start our queries

social-analyzer --username wireshark

As you can see we are now searching many different sites!

```
tsurugi@tsurugi:~$ social-analyzer --username wireshark
[init] Detections are updated very often, make sure to get the most up-to-date o
nes
[init] languages.json looks good!
[init] sites.json looks good!
[init] languages.json & sites.json loaded successfully
[Init] Selected websites: 999
[Info] username: wireshark
[Checking] 9gag.com
[Checking] airliners.net
[Checking] 500px.com
[Checking] airbit.com
[Checking] 7cups.com
[Checking] akniga.org
```

For each detected we will have a block of detected

```
----------------------
found        : 3
link         : https://wireshark.tumblr.com
rate         : %100.0
status       : good
title        : @wireshark on Tumblr
language     : English (Maybe)
country      : United States
rank         : 120
text         : @wireshark on Tumblr Explore Communities Change palette Sign up L
og in sharkify @ wireshark work as terminal hydra 🦈 Ask me anything Follow Post
s Likes Following
type         : Computers Electronics and Technology > Social Networks and Online
 Communities
----------------------
```

If you want to write the output of your file to a text file in the Present Working Directory. (wireshark.txt in this case) The only challenge with this is you will not see the progress as it runs.

social-analyzer –username wireshark > wireshark.txt

if you need to abort you can tap ‘Control’ + ‘C’ a few times.

For more advances queries look at the instruction when you launch the app or simply look at the social-analyzer github:
https://github.com/qeeqbox/social-analyzer

theHarvester

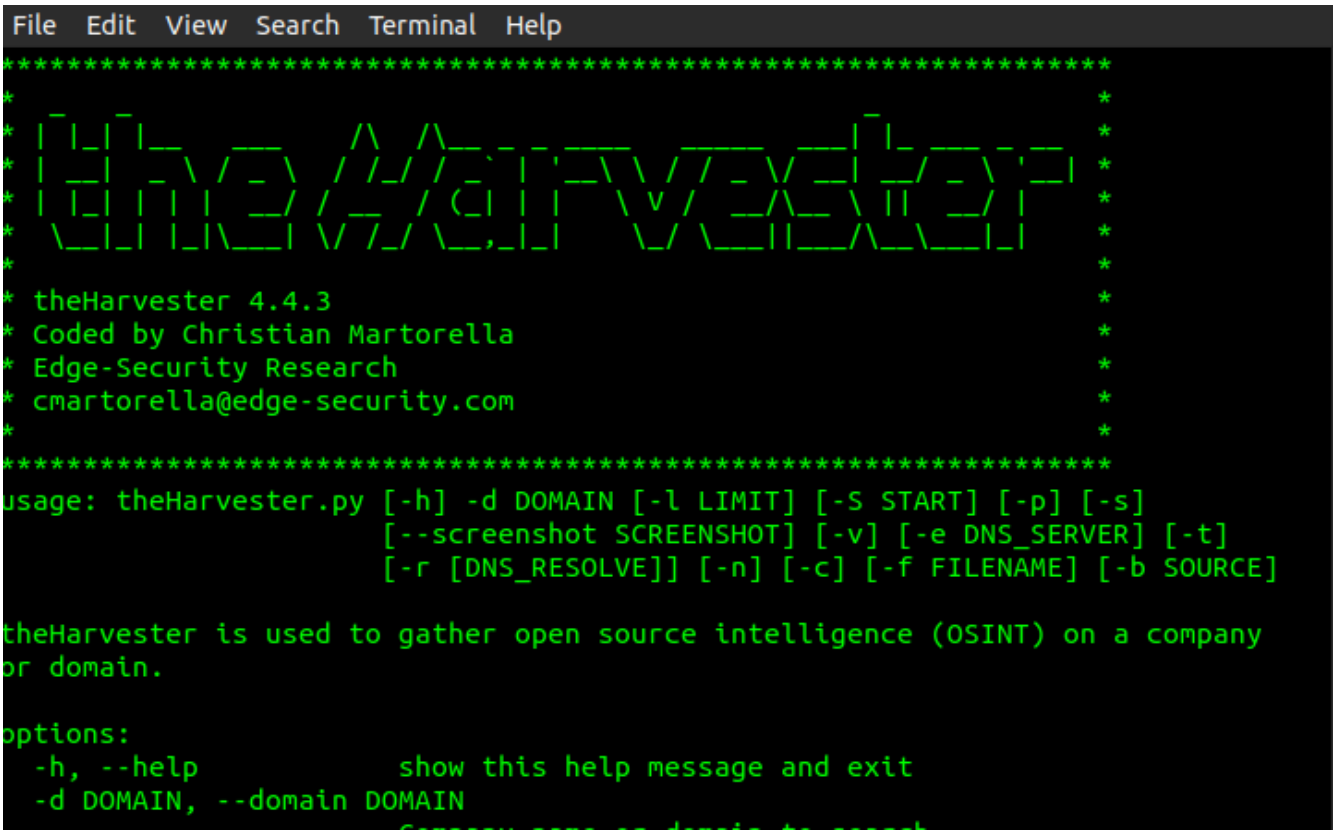
“

theHarvester is a simple to use, yet powerful tool designed to be used during the reconnaissance stage of a red team assessment or penetration test. It performs open source intelligence (OSINT) gathering to help determine a domain's external threat landscape. The tool gathers names, emails, IPs, subdomains, and URLs by using multiple public resources that include:

“

https://github.com/laramies/theHarvester

To open the application click ‘Applications’ in the upper left -> ‘Tsurugi’ -> ‘OSINT’ -> 'Web' -> 'The Harvester'



Once again we get good documentation of how to use the application!

Lets run a simple scan:

theHarvester.py -d kali.org -l 500 -b yahoo -f harvester.txt

-l sets the limit
-b set the source host to use
-f saves the output to the filename

```
tsurugi@tsurugi:~$ theHarvester.py -d kali.org -l 500 -b yahoo -f harvister.txt
*******************************************************************
*                                                                 *
*  _   _                                            _             *
* | |_| |__   ___    /\  /\__ _ _ ____   _____  ___| |_ ___ _ __  *
* | __|  _ \ / _ \  / /_/ / _` | '__\ \ / / _ \/ __| __/ _ \ '__| *
* | |_| | | |  __/ / __  / (_| | |   \ V /  __/\__ \ ||  __/ |    *
*  \__|_| |_|\___| \/ /_/ \__,_|_|    \_/ \___||___/\__\___|_|    *
*                                                                 *
* theHarvester 4.4.3                                              *
* Coded by Christian Martorella                                   *
* Edge-Security Research                                          *
* cmartorella@edge-security.com                                   *
*                                                                 *
*******************************************************************

[*] Target: kali.org
```

```
[*] Target: kali.org

[*] Searching Yahoo.

[*] No IPs found.

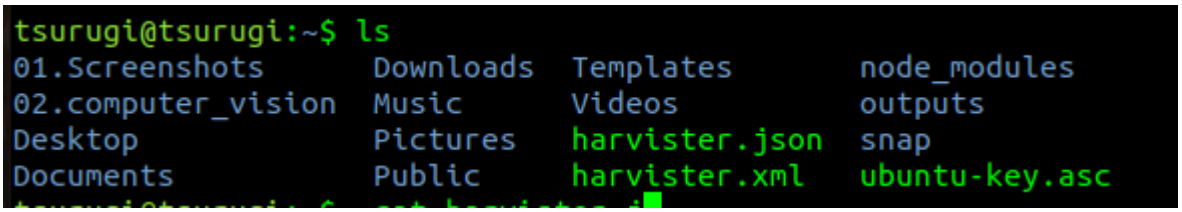
[*] No emails found.

[*] Hosts found: 9
---------------------
archive.kali.org
autopkgtest.kali.org
bugs.kali.org
cdimage.kali.org
http.kali.org
nethunter.kali.org
old.kali.org
pkg.kali.org
status.kali.org

[*] Reporting started.
[*] XML File saved.
[*] JSON File saved.
```

*during the writing of this guide I found that I ran into a bug where I could not use google to perform a search, this might be due to google thinking I am a bot based on previous searches

You can see that the output of this command was saved to both .json and .xml



Of course for more advanced scan check out the help file or theHarvester github:

https://github.com/laramies/theHarvester

Intelligence X

> “
> Intelligence X is a search engine and data archive.
> Intelligence X is an independent European technology company founded in 2018 by Peter Kleissner. The company is based in Prague, Czech Republic. Its mission is to develop and maintain the search engine and data archive.
> Intelligence X differentiates itself from other search engines in these unique ways:
>
> 1. The search works with selectors, i.e. specific search terms such as email addresses, domains, URLs, IPs, CIDRs, Bitcoin addresses, IPFS hashes, etc.
> 2. It searches in places such as the darknet, document sharing platforms, whois data, public data leaks and others.
> 3. It keeps a historical data archive of results, similar to how the Wayback Machine from archive.org stores historical copies of websites.
>
> Our target customers are companies of all sizes and governments. Contact us for individual plans that match your organizations' needs.
> You can use Intelligence X to perform any kind of open source intelligence. We deliver fast, high-quality results and make the deepest parts of the internet accessible with a few clicks. Intelligence X searches billions of selectors in a matter of milliseconds. Combined with our data archive this is a powerful new tool. Here are some examples of Intelligence X's power:
>
> - Research the Trump Organization's email server reported by ArsTechnica
> - Research the Austrian Parliament
> - Investigate a Bitcoin address
> - Investigate a Social Security Number
> - List all .com email addresses on North Korean websites
> - Read the Hillary emails (FOIA/UNCLASSIFIED)
>
> “

https://intelx.io/

Open up Firefox and do a quick web searchto ensure your host can connect to the internet..

‘Applications’ -> ‘Internet’ -> ‘Firefox’

Do a few search of your choosing.

Until Next time!